



AI INNOVATION HACKATHON 2026

Biomedical AI — Image & Signal Analysis

STANDARDIZED PROBLEM STATEMENTS

Women in AI, Research, Innovation & Entrepreneurship Club

Research • Innovation • Entrepreneurship

5 Challenges • Online • AI Innovation Hackathon 2026

Each challenge follows the same structured format for clarity, consistency and ease of review.

CHALLENGE PORTFOLIO

NO.	CHALLENGE	DIFFICULTY
1	AI-Powered Multi-Disease Chest X-Ray Screening Assistant	Intermediate
2	AI-Based Skin Lesion Risk Analysis & Explainable Image Assistant	Intermediate
3	AI ECG Rhythm Monitoring & Abnormality Detection System	Intermediate
4	AI-Based Sleep Apnea Detection from Physiological Signals	Intermediate-Advanced
5	AI Brain MRI Tumor Segmentation & Visualization	Advanced



CHALLENGE 01

AI-Powered Multi-Disease Chest X-Ray Screening Assistant

DOMAIN

Medical Imaging

DIFFICULTY

Intermediate

PROBLEM STATEMENT

Chest X-rays are routinely used to assess a wide range of thoracic conditions. Reviewing large numbers of images can be time-consuming, while some abnormalities may be subtle. Participants are challenged to develop an AI system that analyzes a chest X-ray and identifies one or more predefined abnormalities.

EXPECTED SOLUTION

Build a prototype in which a participant uploads a chest X-ray, the model processes the image, and the interface displays the predicted class or classes with confidence scores. Teams may optionally provide an explanation map that highlights image regions contributing to the prediction. The prototype should be explicitly presented as an educational/research screening system rather than a clinical diagnostic device.

SUGGESTED DATASETS / RESOURCES

NIH ChestX-ray14; CheXpert; MIMIC-CXR; VinDr-CXR. Follow the licensing, access, privacy and usage requirements of the selected dataset.

AI / ML COMPONENT

Transfer learning with CNNs; multi-label image classification; image preprocessing and augmentation; optional Grad-CAM/explainable AI; optional abnormal-region localization.

WHAT MAKES THE CHALLENGE INTERESTING?

The challenge produces a highly visible end-to-end AI demonstration and can be extended into explainability, dataset-bias analysis, model comparison and responsible medical AI evaluation.

SUGGESTED ON-SCREEN DEMONSTRATION

Upload X-ray AI inference predicted findings confidence scores optional highlighted regions.

TECHNICAL REQUIREMENTS

Python; image preprocessing; CNN/transfer learning; multi-label evaluation; explainability; simple web/notebook interface.



SUGGESTED MINIMUM DELIVERABLE

- Working prototype with reproducible source code/notebook.
- README describing the problem, data, methodology, implementation and limitations.
- Demonstration of the core AI/ML functionality on representative inputs.
- Evaluation results using appropriate metrics for the selected task.
- Provide predicted finding classes with confidence scores.
- Include an optional or required visual explanation depending on team approach.

OPTIONAL ADVANCED FEATURES

- Add abnormal-region localization.
- Compare multiple pretrained backbones.
- Perform subgroup/dataset-bias analysis.
- Add calibration or uncertainty analysis.

RESPONSIBLE AI / ETHICAL CONSIDERATIONS

All challenges should be framed as research/educational prototypes, not clinical diagnostic or treatment systems. Participants should document dataset limitations, avoid unsupported clinical claims, respect dataset licensing and privacy requirements, and report relevant bias, class imbalance, representation and validation limitations.

INNOVATION ANGLE

Combines multi-label screening, visual explanation and responsible evaluation in one demonstrable prototype.

SUCCESS / EVALUATION METRICS

Per-class precision/recall/F1, AUROC where appropriate, calibration/confidence quality, explanation quality and error analysis.



CHALLENGE 02

AI-Based Skin Lesion Risk Analysis & Explainable Image Assistant

DOMAIN

Medical Imaging

DIFFICULTY

Intermediate

PROBLEM STATEMENT

Skin lesions can have visually similar characteristics, making automated image analysis a challenging computer-vision problem. Participants are challenged to classify lesion images into predefined categories and provide an interpretable visual explanation of the model's prediction.

EXPECTED SOLUTION

Create a prototype where a user uploads a lesion image and receives a predicted category and confidence score. The interface should optionally display a heatmap or attention map indicating which image regions influenced the model. The result should be framed as an AI research/educational tool, not a medical diagnosis.

SUGGESTED DATASETS / RESOURCES

ISIC Archive datasets, including ISIC 2018/2019/2020 challenge data; HAM10000; other officially provided ISIC resources. Verify current dataset terms and permitted uses.

AI / ML COMPONENT

CNN/transfer learning; image classification; data augmentation; optional lesion segmentation; Grad-CAM or other explainability methods; optional multimodal features such as age or anatomical site where legally and ethically appropriate.

WHAT MAKES THE CHALLENGE INTERESTING?

Connects computer vision with a visible healthcare use case and supports future work in explainable AI, telehealth-oriented research, multimodal modelling, dataset diversity and responsible AI.

SUGGESTED ON-SCREEN DEMONSTRATION

Upload skin image AI classification confidence score optional lesion/attention heatmap.

TECHNICAL REQUIREMENTS

Python; image preprocessing; classification; explainability; visualization/interface; dataset documentation.

SUGGESTED MINIMUM DELIVERABLE

- Working prototype with reproducible source code/notebook.
- README describing the problem, data, methodology, implementation and limitations.
- Demonstration of the core AI/ML functionality on representative inputs.
- Evaluation results using appropriate metrics for the selected task.
- Predict predefined categories with confidence.
- Provide an interpretable image-region explanation.

OPTIONAL ADVANCED FEATURES

- Add lesion segmentation.
- Compare explainability methods.
- Investigate dataset diversity or subgroup performance.
- Explore carefully governed multimodal features.

RESPONSIBLE AI / ETHICAL CONSIDERATIONS



All challenges should be framed as research/educational prototypes, not clinical diagnostic or treatment systems. Participants should document dataset limitations, avoid unsupported clinical claims, respect dataset licensing and privacy requirements, and report relevant bias, class imbalance, representation and validation limitations.

INNOVATION ANGLE

Combines image classification with visual explanation and explicit attention to dataset diversity and responsible use.

SUCCESS / EVALUATION METRICS

Per-class classification metrics, calibration/confidence, explanation consistency, subgroup performance where available, and error analysis.



CHALLENGE 03

AI ECG Rhythm Monitoring & Abnormality Detection System

DOMAIN

Biomedical Signal Processing

DIFFICULTY

Intermediate

PROBLEM STATEMENT

ECG recordings contain temporal patterns that can indicate different cardiac rhythms. Long recordings may contain many beats and segments, making automated analysis a useful signal-processing challenge. Participants are asked to classify selected ECG segments or beats into predefined rhythm/heartbeat categories.

EXPECTED SOLUTION

Develop a prototype that displays an ECG waveform, applies basic preprocessing, segments the signal where appropriate, and produces an AI-based rhythm classification. The interface should visually show the analyzed signal and the model's output, with abnormal or selected segments optionally highlighted.

SUGGESTED DATASETS / RESOURCES

MIT-BIH Arrhythmia Database via PhysioNet; MIT-BIH Normal Sinus Rhythm Database; MIT-BIH Atrial Fibrillation Database.

AI / ML COMPONENT

Signal preprocessing and filtering; feature extraction; 1D-CNN; recurrent neural networks/LSTM; time-series classification; optional Transformer-based sequence modelling.

WHAT MAKES THE CHALLENGE INTERESTING?

Ensures genuine biomedical signal processing rather than only image classification and provides a highly demonstrable waveform-to-prediction workflow.

SUGGESTED ON-SCREEN DEMONSTRATION

Load ECG waveform display preprocessing selected segment classification highlighted output.

TECHNICAL REQUIREMENTS

Python; signal preprocessing; time-series segmentation; classification; waveform visualization; reproducible evaluation.

SUGGESTED MINIMUM DELIVERABLE

- Working prototype with reproducible source code/notebook.
- README describing the problem, data, methodology, implementation and limitations.
- Demonstration of the core AI/ML functionality on representative inputs.
- Evaluation results using appropriate metrics for the selected task.
- Display an ECG waveform and analyzed segment.
- Produce rhythm classification with appropriate evaluation.

OPTIONAL ADVANCED FEATURES

- Compare 1D-CNN, LSTM and/or Transformer approaches.
- Add beat-level or event-level highlighting.
- Explore multimodal physiological-signal fusion where permitted.

RESPONSIBLE AI / ETHICAL CONSIDERATIONS



All challenges should be framed as research/educational prototypes, not clinical diagnostic or treatment systems. Participants should document dataset limitations, avoid unsupported clinical claims, respect dataset licensing and privacy requirements, and report relevant bias, class imbalance, representation and validation limitations.

INNOVATION ANGLE

Combines interpretable waveform visualization with AI-based temporal classification.

SUCCESS / EVALUATION METRICS

Per-class precision/recall/F1, sensitivity/specificity where appropriate, segment-level or beat-level error analysis, and robustness to preprocessing choices.



CHALLENGE 04

AI-Based Sleep Apnea Detection from Physiological Signals

DOMAIN

Biomedical Signal Processing

DIFFICULTY

Intermediate–Advanced

PROBLEM STATEMENT

Sleep-related breathing disturbances produce patterns that can be studied through physiological signals recorded during sleep. Participants are challenged to analyze ECG and/or other available physiological signals and identify time periods associated with apnea events.

EXPECTED SOLUTION

Build a system that takes a physiological recording, preprocesses the signal, predicts apnea/non-apnea periods or selected event categories, and presents the result as a timeline. A strong prototype can overlay predicted events on the original signal so that the model's output can be inspected visually.

SUGGESTED DATASETS / RESOURCES

PhysioNet Apnea-ECG Database; MIT-BIH Polysomnographic Database. Use only signals and annotations permitted by the selected dataset and clearly document preprocessing and evaluation.

AI / ML COMPONENT

Biomedical signal processing; time-series classification; 1D-CNN; LSTM/Transformer models; event detection; optional multimodal physiological-signal fusion.

WHAT MAKES THE CHALLENGE INTERESTING?

Moves beyond single-image prediction into continuous physiological time-series analysis with potential relevance to wearable-health research and remote monitoring.

SUGGESTED ON-SCREEN DEMONSTRATION

Load overnight physiological signal timeline visualization AI event detection apnea periods highlighted.

TECHNICAL REQUIREMENTS

Python; physiological signal preprocessing; event detection; time-series modelling; timeline visualization; evaluation.

SUGGESTED MINIMUM DELIVERABLE

- Working prototype with reproducible source code/notebook.
- README describing the problem, data, methodology, implementation and limitations.
- Demonstration of the core AI/ML functionality on representative inputs.
- Evaluation results using appropriate metrics for the selected task.
- Predict apnea/non-apnea periods or selected event categories.
- Present predictions on a timeline aligned with the input signal.

OPTIONAL ADVANCED FEATURES

- Add multimodal signal fusion.
- Compare event-detection strategies.
- Add uncertainty estimates or threshold analysis.
- Evaluate robustness across permitted recordings.



RESPONSIBLE AI / ETHICAL CONSIDERATIONS

All challenges should be framed as research/educational prototypes, not clinical diagnostic or treatment systems. Participants should document dataset limitations, avoid unsupported clinical claims, respect dataset licensing and privacy requirements, and report relevant bias, class imbalance, representation and validation limitations.

INNOVATION ANGLE

Continuous signal/event analysis with a visible timeline rather than only a static classification output.

SUCCESS / EVALUATION METRICS

Event detection precision/recall/F1, temporal overlap or event-timing error, robustness across recordings, and calibration where used.



CHALLENGE 05

AI Brain MRI Tumor Segmentation & Visualization

DOMAIN

Medical Imaging

DIFFICULTY

Advanced

PROBLEM STATEMENT

Brain MRI scans contain complex anatomical structures, and identifying the spatial extent of a tumor is a challenging medical-image analysis task. Participants are challenged to develop an AI model that identifies and segments predefined tumor regions in MRI data.

EXPECTED SOLUTION

Develop a prototype that accepts an MRI image or selected MRI slice/volume, generates a predicted segmentation mask, and overlays the predicted tumor region on the original MRI. Teams should report an appropriate segmentation metric and visually inspect prediction quality.

SUGGESTED DATASETS / RESOURCES

BraTS challenge datasets and officially provided BraTS resources. Follow current challenge rules, dataset access conditions and data-use requirements.

AI / ML COMPONENT

Medical-image preprocessing; U-Net or related segmentation architectures; semantic segmentation; Dice score/IoU evaluation; optional 3D CNNs and 3D visualization.

WHAT MAKES THE CHALLENGE INTERESTING?

The challenge asks AI to identify not only whether an abnormality exists but also where it is. The segmentation overlay creates a strong visual demonstration and introduces participants to a major research area in medical computer vision.

SUGGESTED ON-SCREEN DEMONSTRATION

Load MRI slice/volume AI segmentation tumor mask segmentation overlay metric/result display.

TECHNICAL REQUIREMENTS

Python; medical-image preprocessing; segmentation; mask visualization; Dice/IoU evaluation; reproducible pipeline.

SUGGESTED MINIMUM DELIVERABLE

- Working prototype with reproducible source code/notebook.
- README describing the problem, data, methodology, implementation and limitations.
- Demonstration of the core AI/ML functionality on representative inputs.
- Evaluation results using appropriate metrics for the selected task.
- Generate a predicted segmentation mask.
- Overlay the prediction on the MRI and report a segmentation metric.

OPTIONAL ADVANCED FEATURES

- Use 3D segmentation/visualization.
- Compare segmentation architectures.
- Add uncertainty or boundary-quality analysis.
- Perform error analysis across tumor regions where labels permit.

RESPONSIBLE AI / ETHICAL CONSIDERATIONS



All challenges should be framed as research/educational prototypes, not clinical diagnostic or treatment systems. Participants should document dataset limitations, avoid unsupported clinical claims, respect dataset licensing and privacy requirements, and report relevant bias, class imbalance, representation and validation limitations.

INNOVATION ANGLE

Spatially explicit medical-image AI with a directly inspectable segmentation result.

SUCCESS / EVALUATION METRICS

Dice score, IoU, sensitivity/recall for relevant regions, boundary/visual quality, and robustness/error analysis.